Microsoft 365

MD-100 Windows Client

Module 9: Support the Windows client environment

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Module Agenda

Explore troubleshooting methodologies

Explore Windows architecture

Explore support and diagnostic tools

Monitor and troubleshoot Windows client performance

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Lesson 1: Explore troubleshooting methodologies

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Lesson 1: Explore troubleshooting methodologies

- Examine the Enterprise Desktop Support Technician role
- Examine the Desktop Support Environment
- Explore key stages and terminology of a troubleshooting methodology
- Examine the process of problem reporting

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Examine the Enterprise Desktop Support Technician role

- Good troubleshooter
- Knowledge resource
- Effective communicator
- Informative source

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Examine the Desktop Support Environment

- Workgroups
  - Peer-to-peer
- Domains
  - logical groups of networked devices
- Cloud-based infrastructure
  - infrastructure hosted through a cloud service

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Explore key stages and terminology of a troubleshooting methodology

Common Processes:

- Classify
- Test
- Escalation
- Report

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Examine the process of problem reporting

- Detect problems
- Encourage end-users to help themselves
  - Provide training, self-service resources
- Contact the help desk
- Document and classify
- Escalate problems as needed
- Resolve the problem
- Close the problem

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Knowledge Check

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Lesson 2: Explore Windows architecture

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Lesson 2: Explore Windows architecture

- Compare Windows client devices
- Examine the Windows client architecture
- Examine the Desktop Support Environment
- Explore key stages and terminology of a troubleshooting methodology

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Compare Windows client devices

Desktop computers

Laptop computers

Convertible laptops

Tablets

Device selection factors

- Battery life
- Processor performance
- Screen size/resolution
- External expansion options
- Memory
- Storage

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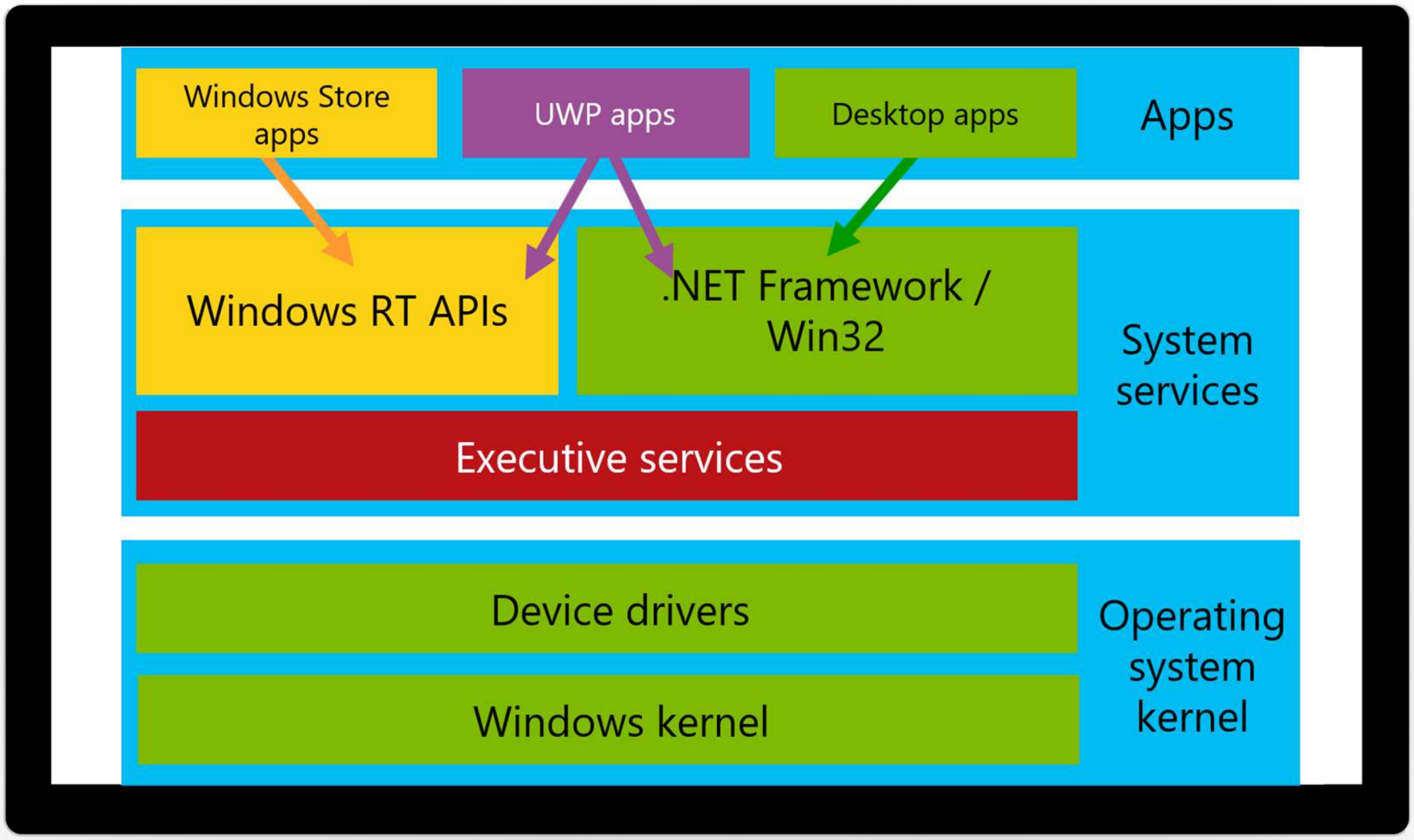
Examine the Windows client architecture



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Examine the Desktop Support Environment

- Workgroups
- Domains
- Cloud-based services

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Explore key stages and terminology of a troubleshooting methodology

- Classification
- Testing
- Escalation
- Reporting

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Lesson 3: Explore support and diagnostic tools

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Lesson 3: Explore support and diagnostic tools

- Examine Task Manager
- Examine Resource Monitor
- Examine Event Viewer
- Explain Performance Monitor
- Explain Reliability Monitor
- Compare Process Explorer and Process Monitor
- Examine the Diagnostics and Recovery Toolset
- Use the Steps Recorder
- Explore the Microsoft Management Console
- Examine the Windows registry
- Use the Windows registry editor
- Explore additional tools

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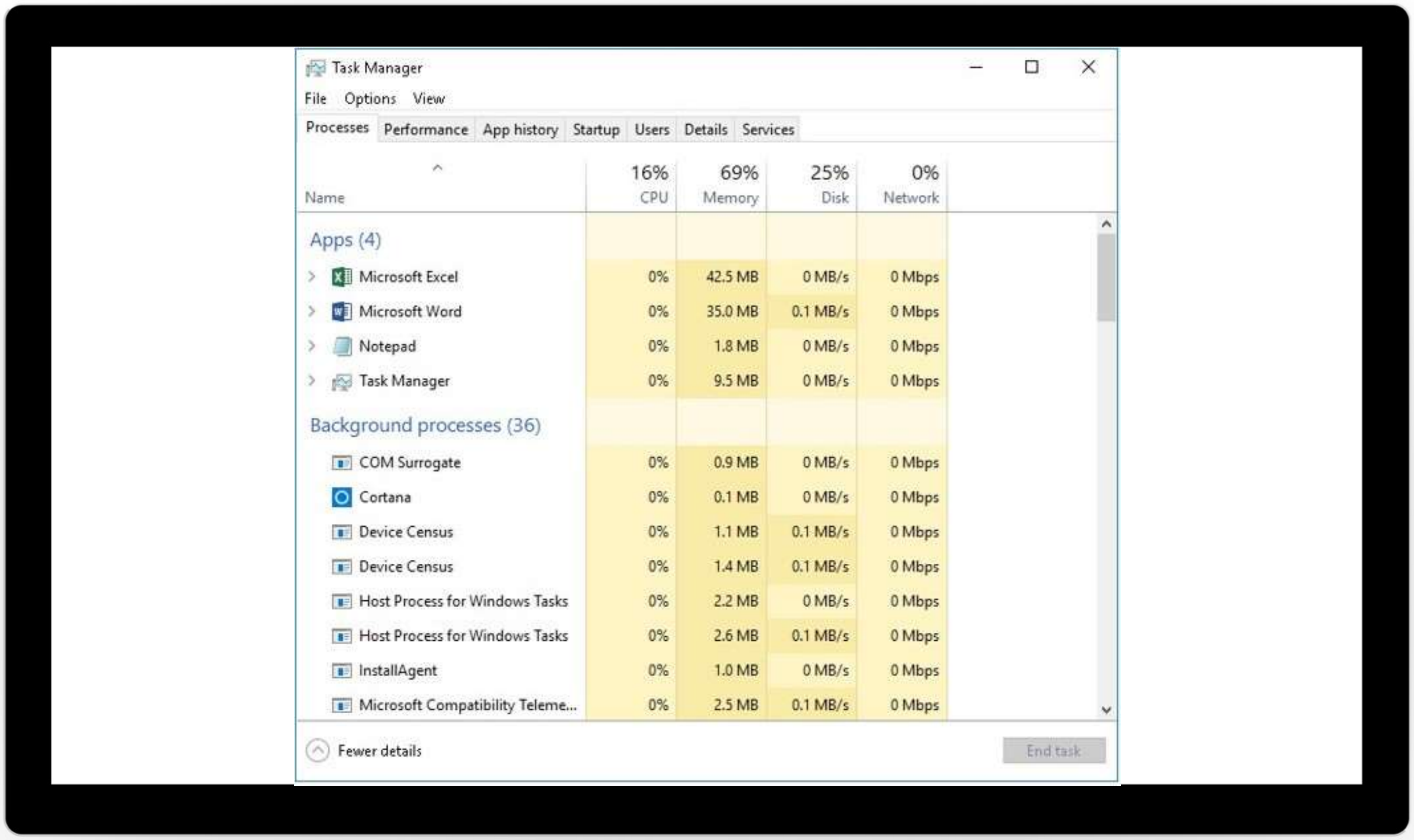
Examine Task Manager



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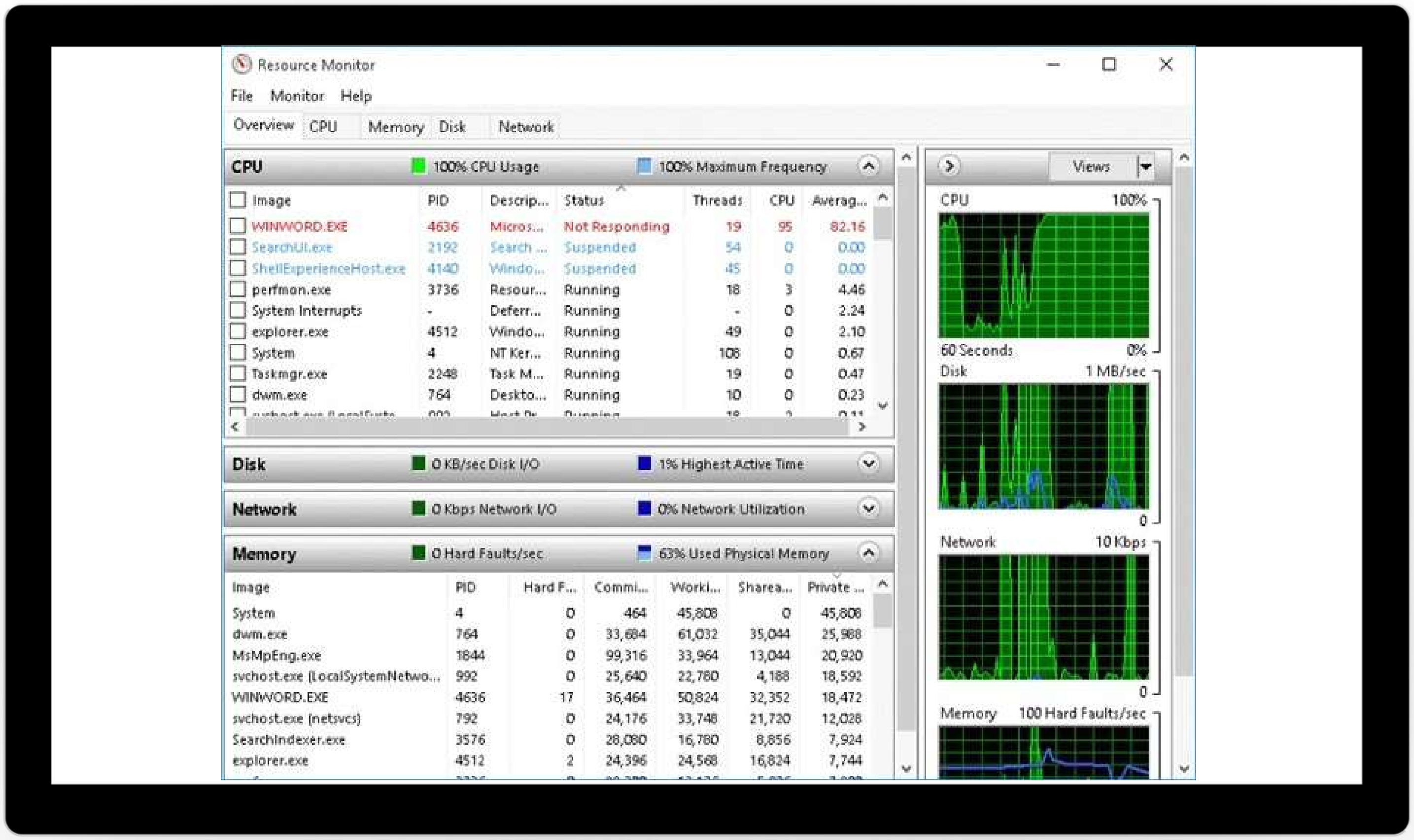
Examine Resource Monitor



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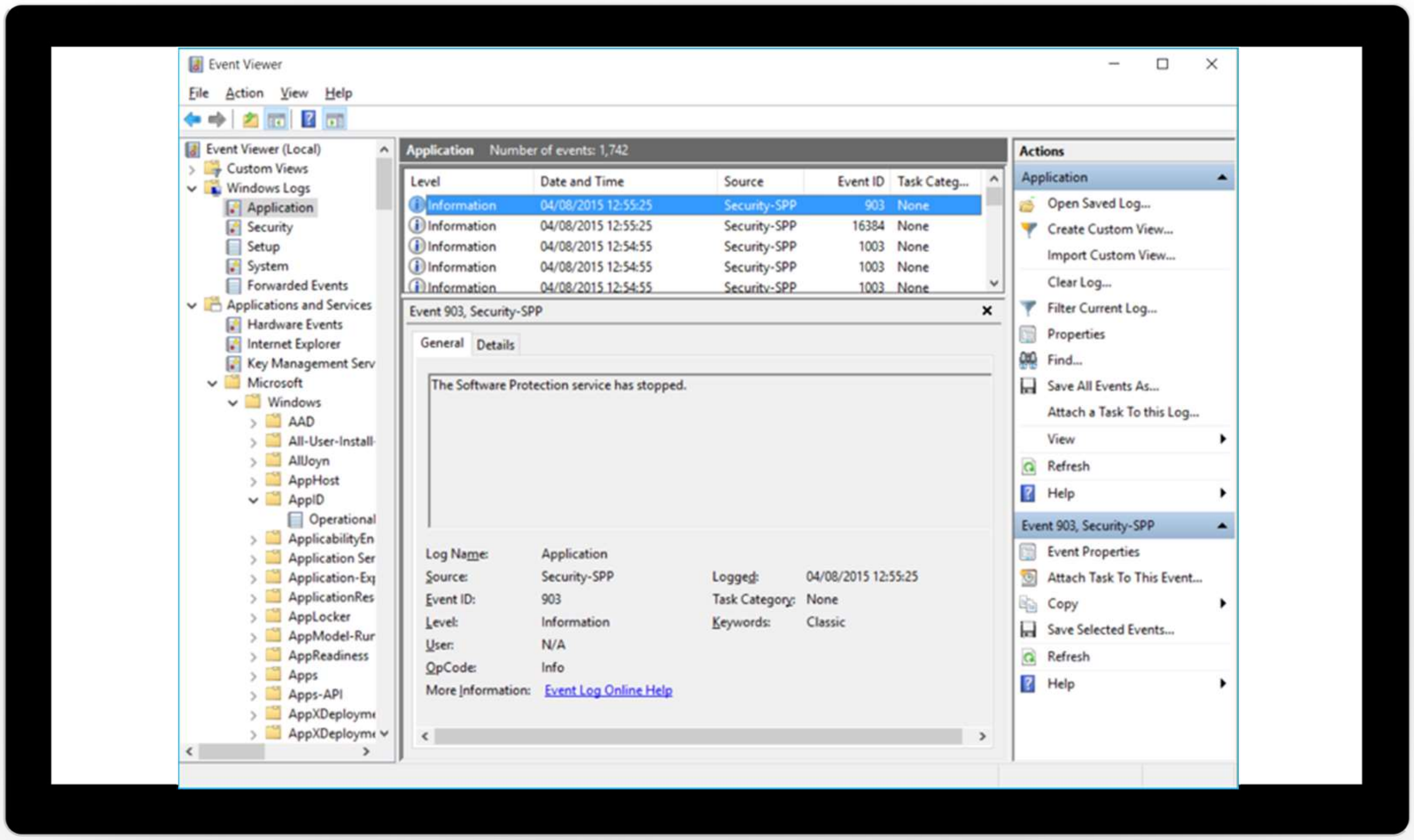
Examine Event Viewer



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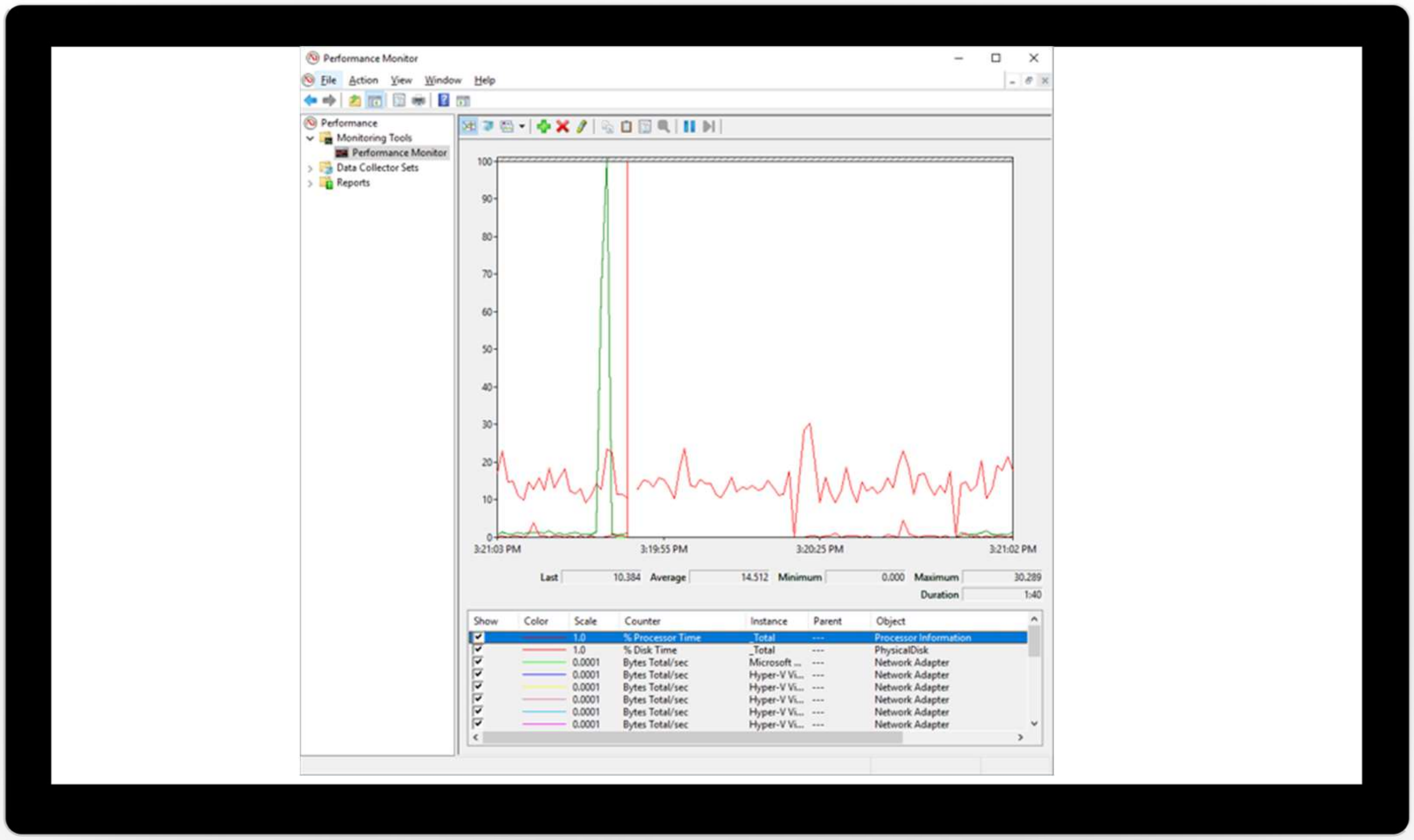
Explain Performance Monitor



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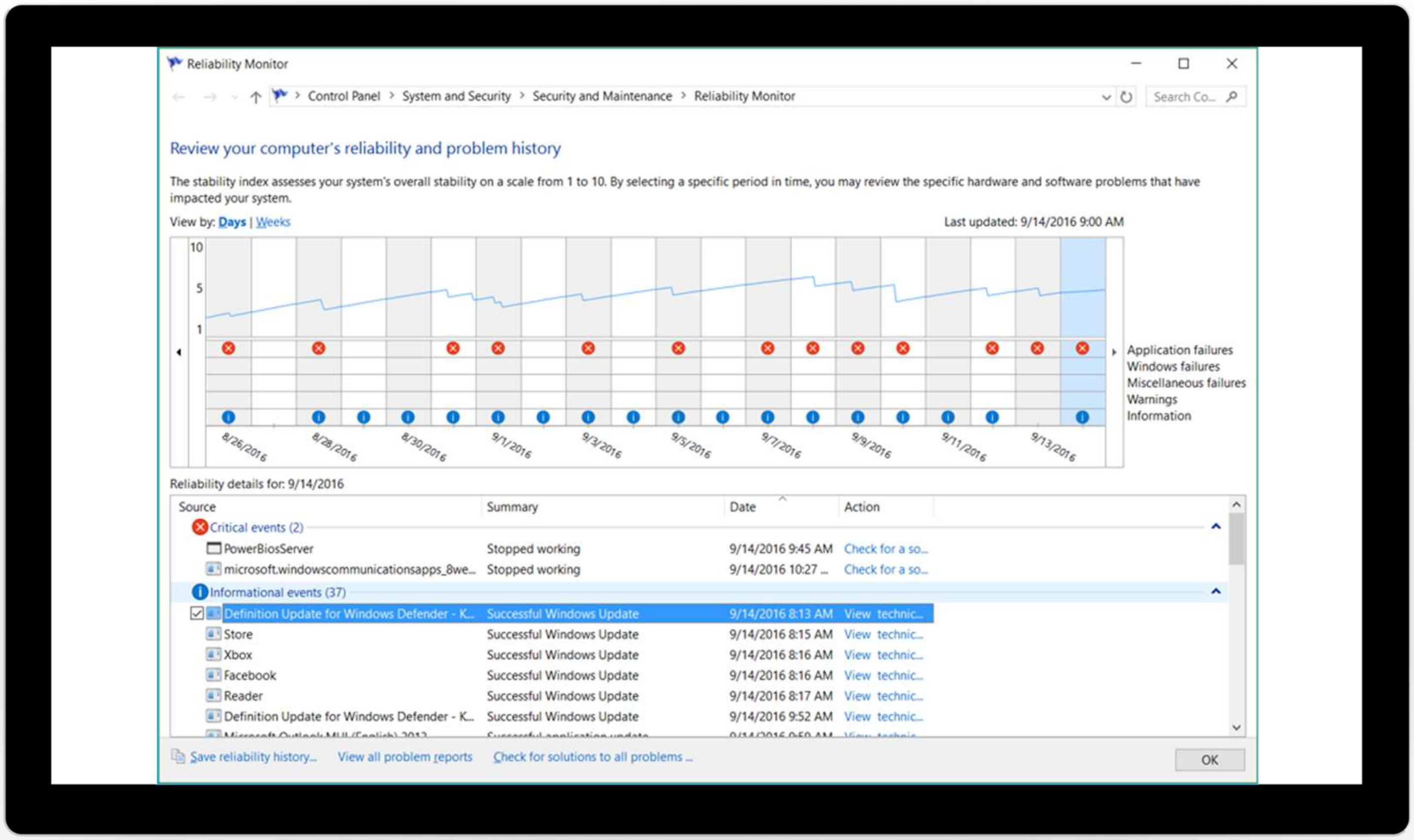
Explain Reliability Monitor



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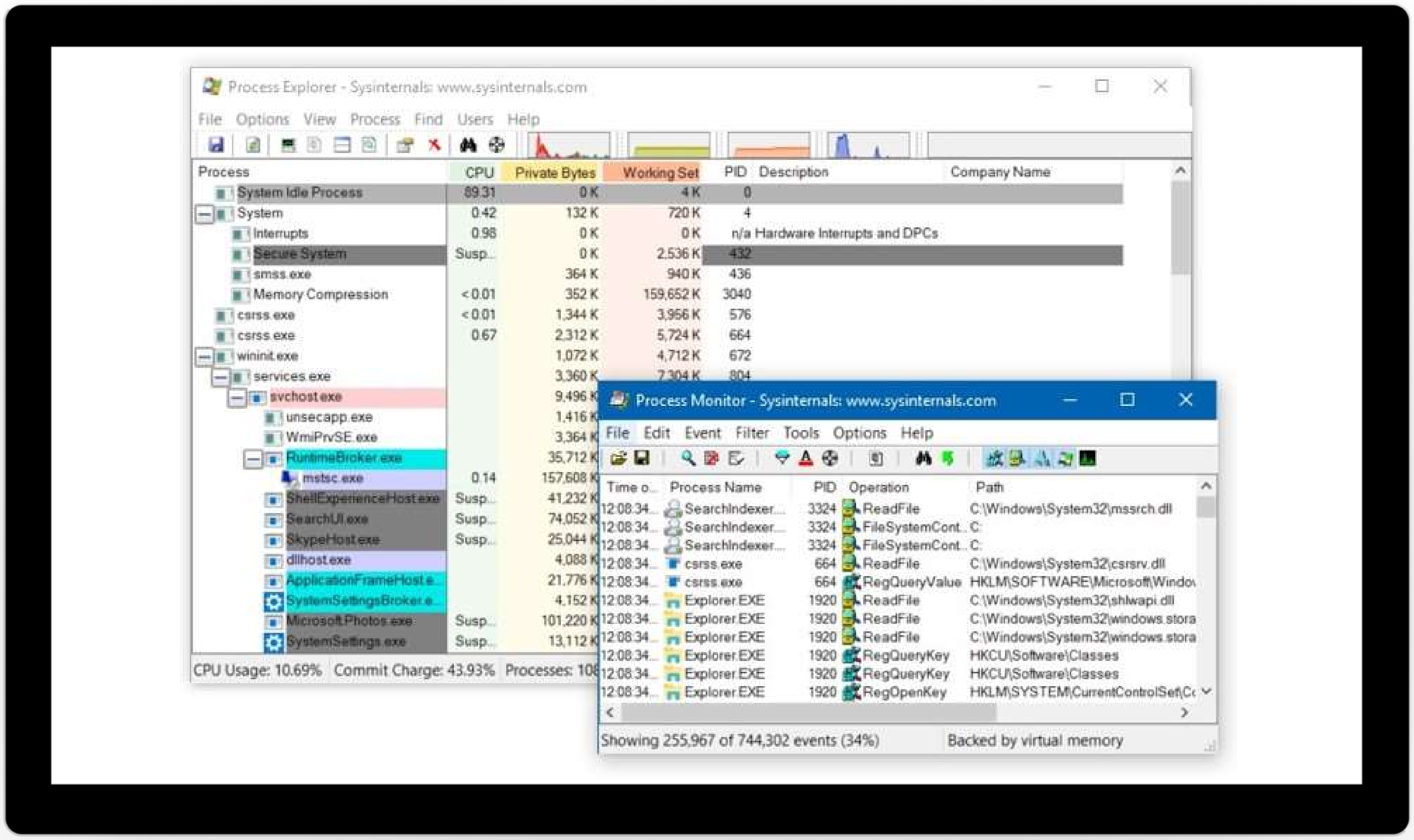
Compare Process Explorer and Process Monitor



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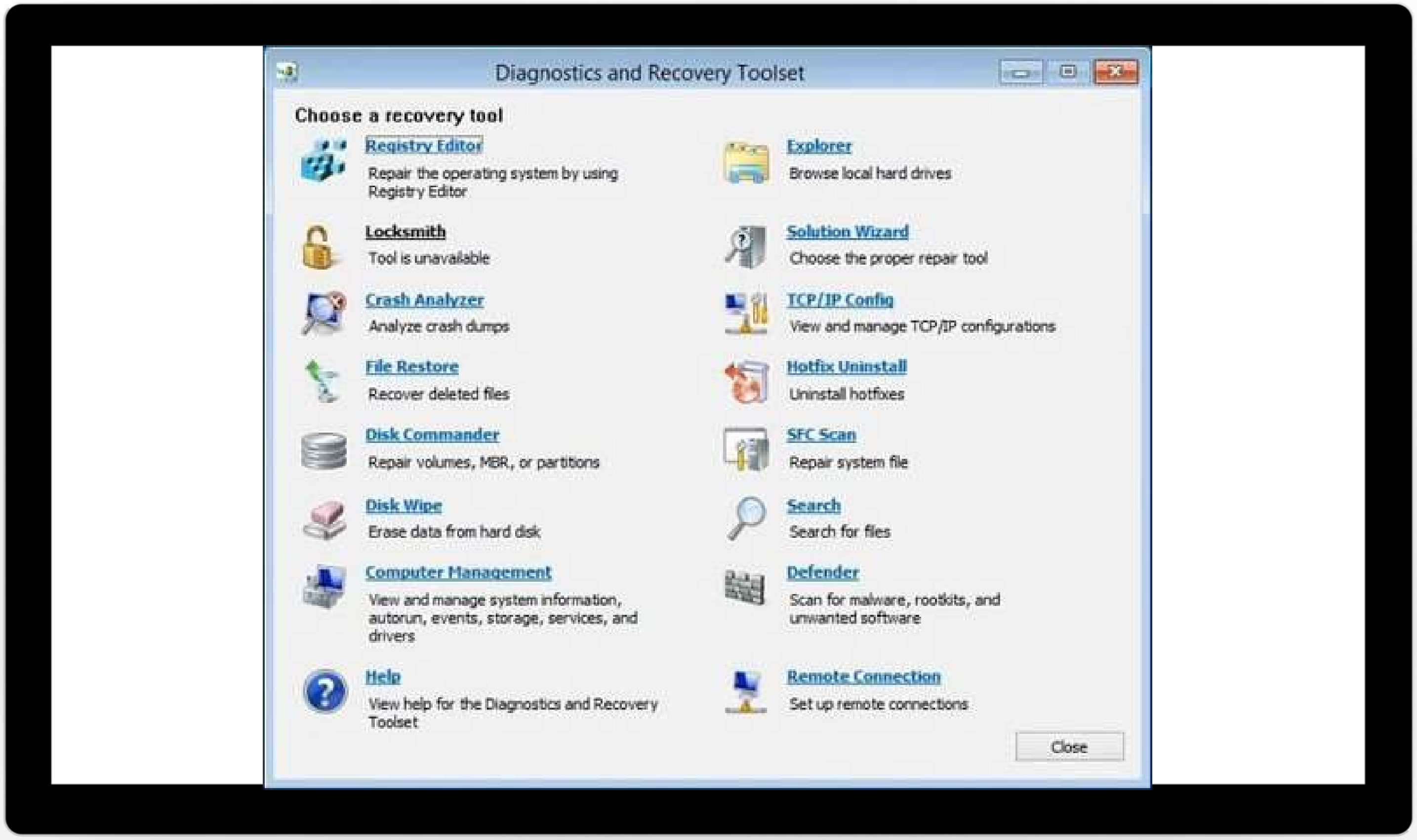
Examine the Diagnostics and Recovery Toolset



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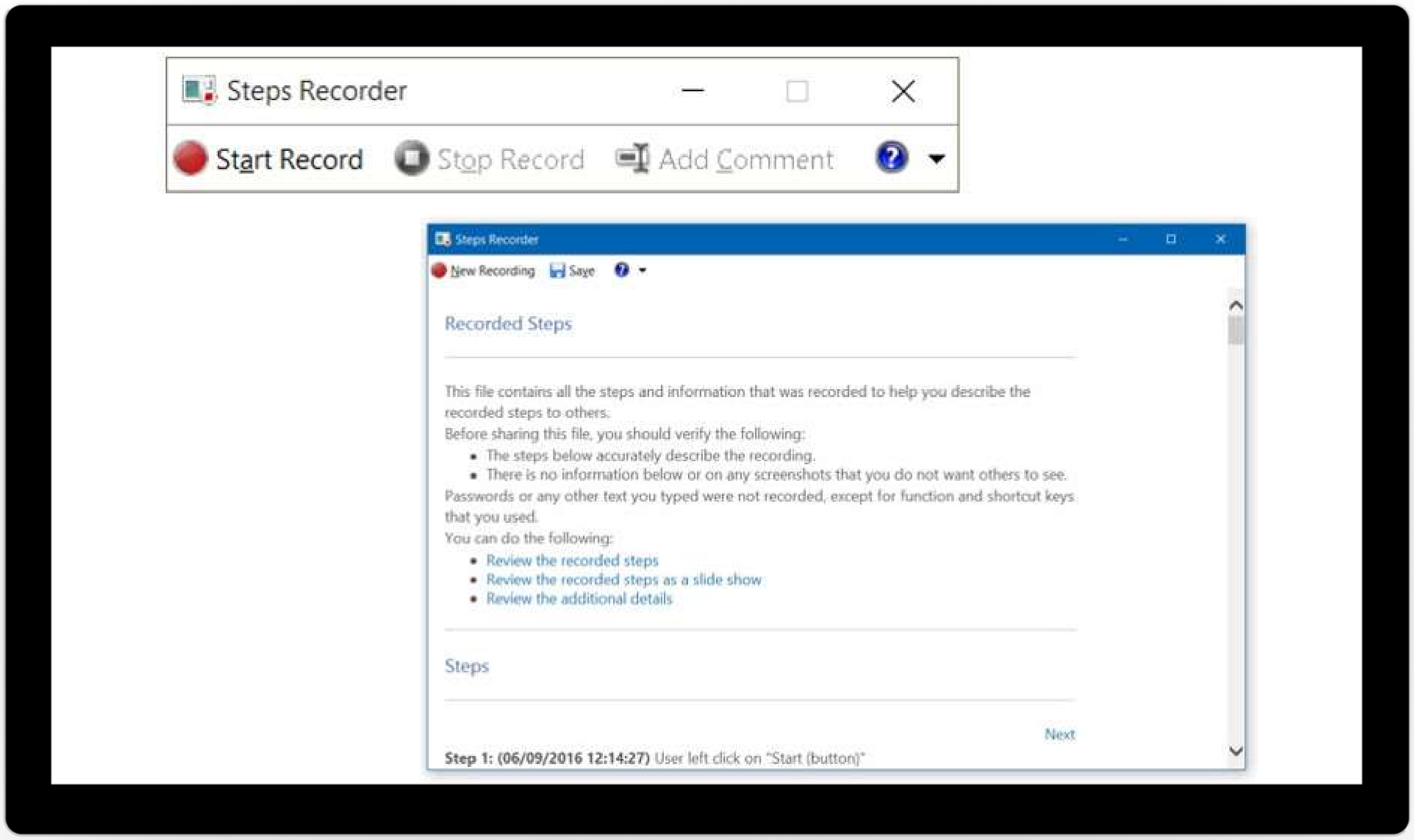
Use the Steps Recorder



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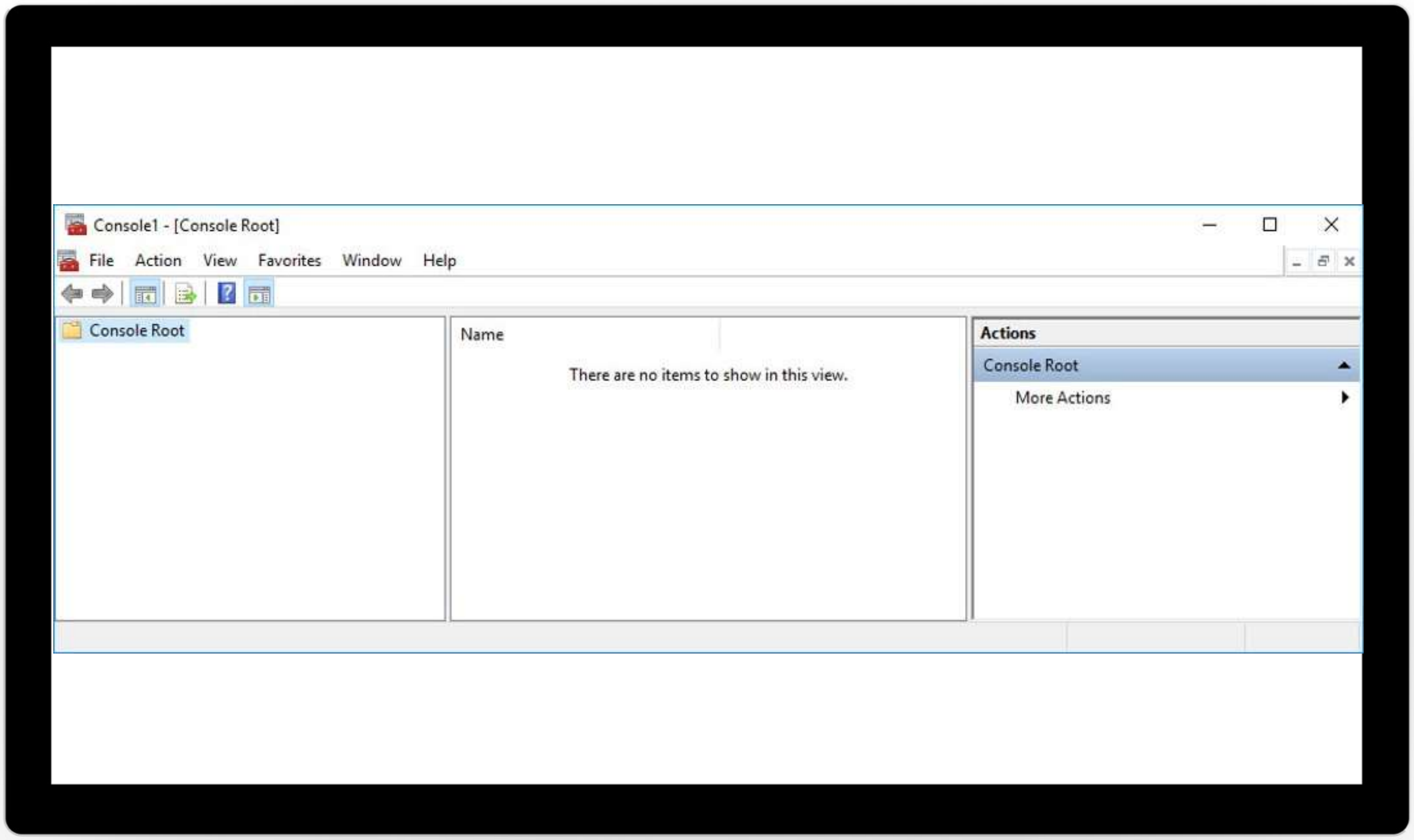
Explore the Microsoft Management Console



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Examine the Windows registry

Hive	Description
HKEY_CLASSES_ROOT	Contains file association information and defines which application opens
HKEY_CURRENT_USER	Configuration information for the currently signed-in user
HKEY_LOCAL_MACHINE	Computer-related configuration settings
HKEY_USERS	Information about the current hardware profilefiguration information for all users that have signed in locally
HKEY_CURRENT_CONFIG	Information about the current hardware profile

Value type	Data type
REG_BINARY	Binary
REG_DWORD	DWORD
REG_SZ	String
REG_EXPAND_SZ	Expandable string
REG_MULTI_SZ	Multiple strings

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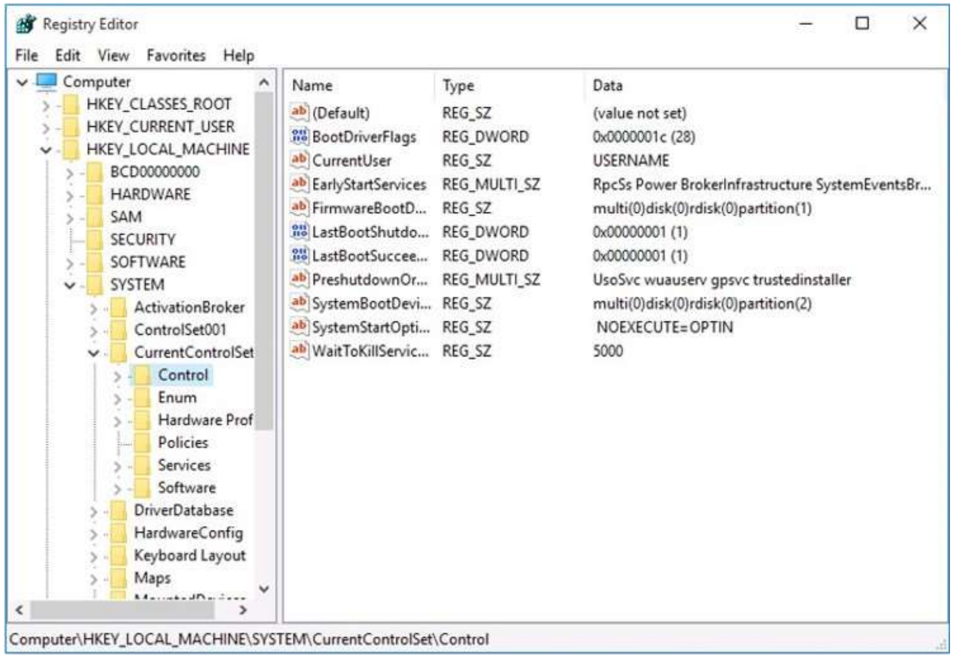
Use the Windows registry editor

- RegEdit
- Windows PowerShell
- Group Policy



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Explore additional tools

- Sysinternals
- Windows Performance Toolkit

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Knowledge Check

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Lesson 4: Monitor and troubleshoot Windows client performance

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Lesson 4: Monitor and troubleshoot Windows client performance

- Examine key performance components in Windows client
- Establish a performance baseline
- Monitor Windows client performance
- Optimize disk and memory performance
- Configure Windows client search indexes

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Examine key performance components in Windows client

- You should monitor these four main hardware components on a Windows client device:
  - Processor
  - Disk
  - Memory
  - Network
- A performance bottleneck occurs when a computer is unable to service the current requests for a specific resource

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Establish a performance baseline

You can configure a performance baseline to help you with:

- Evaluating your computer’s workload
- Monitoring system resources
- Noticing changes and trends in resource use
- Testing configuration changes
- Diagnosing problems

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Monitor Windows client performance

Key Tools:

- Task Manager
- Resource Monitor
- Performance Monitor

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Optimize disk and memory performance

- Use 64-bit
- Dedicated video memory
- Optimize paging
- High-performance storage (SSDs, M.2)
- Enable write-caching
- Distribute workload across available disks

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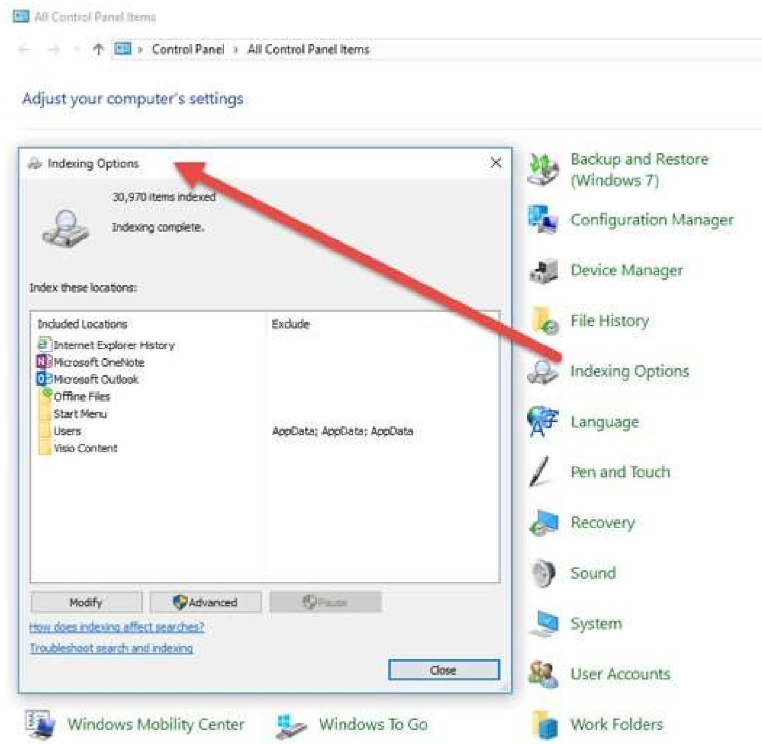
Configure Windows client search indexes

- Define which locations to include or exclude
- Can index encrypted files
- Settings can be applied using Group Policy



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Knowledge Check

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Practice Labs

Monitoring Events

Monitoring Events using Event Viewer

Monitoring Reliability and Performance

Modifying Registry Settings Using Registry Editor

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Module Review

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